Differential equations

algebraic equations:

$$f(x) = 0$$

e.g. $f(x) = x^2 + 3x + 1$

differential equation

$$f\left(t, y(t), \frac{dy(t)}{dt}, \frac{d^2y(t)}{dt^2}, \ldots\right) = 0$$

$y(t)$ is a function of t.

$$\left(\frac{dy}{dx}\right)^2 + 3\left(\frac{dy}{dx}\right) + y + e^{-t} = 0$$

$$y_0(t) \rightarrow y_0(t) + C$$

initial conditions
boundary conditions
$\Big\} \rightarrow y(t_0) = y_0$

First order differential equations

$$\frac{dy}{dt} = f(t, y)$$

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$$\frac{dy}{dt} = f(t, y)$$

separable: $\frac{dy}{dt} = f(y)\,g(t)$

"homogeneous": $\frac{dy}{dt} = f(y/t)$

linear: $\frac{dy}{dt} = -a(t)\,y + f(t)$

"general" $\frac{dy}{dt} = f(t, y)$ numerical

Separable:

$$\frac{dy}{dt} = f(y)\,g(t) \rightarrow \frac{dy}{dt} = -y^2 \sin t$$

($f(y)$: $-y^2$; $g(t)$: $\sin t$)

$$\frac{dy}{f(y)} = g(t)\,dt.$$

$$\int \frac{dy}{f(y)} = \int g(t)\,dt + C \leftarrow y(t_0) = y_0$$

$$\frac{dy}{dt} = -y^2 \sin t$$

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$$\frac{dy}{y^2} = -\sin t\, dt$$

$$\int \frac{dy}{y^2} = -\int \sin t\, dt \leftarrow$$

$$-y^{-1}(t) = \cos t + C \leftarrow y(t_0) = y_0$$

$$-\frac{1}{y(t_0)} = \cos t_0 + C \Rightarrow C = -\frac{1}{y(t_0)} - \cos t_0$$

$$-\frac{1}{y(t)} = \cos t - \frac{1}{y(t_0)} - \cos t_0$$

$$= \cos t - \frac{1}{y_0} - \cos t_0$$

$$\int_{y_0}^{y(t)} \frac{dy}{y^2} = -\int_{t_0}^{t} \sin t\, dt$$

$$-\frac{1}{y}\Big|_{y_0}^{y(t)} = +\cos t\Big|_{t_0}^{t}$$

$$-\frac{1}{y(t)} + \frac{1}{y_0} = \cos t - \cos t_0$$

$$\int_{y_0}^{y(t)} \frac{dy}{f(y)} = \int_{t_0}^{t} g(t)\, dt \leftarrow \frac{dy}{dt} = f(y)g(t)$$

"homogeneous"

$$\frac{dy}{dt} = f(y/t) \rightarrow \frac{dy}{dt} = \frac{t^2 - ty + y^2}{t^2}$$

$$= 1 - \frac{y}{t} + \left(\frac{y}{t}\right)^2$$

$$\uparrow$$

$$f(y/t)$$

$$v(t) = \frac{y(t)}{t}$$

$$y(t) = t\,v(t)$$

$$\frac{dy(t)}{dt} = t\frac{dv}{dt} + v \leftarrow$$

$$\frac{dy}{dt} = f(y/t)$$

$$t\frac{dv}{dt} + v = f(v)$$

$$t\frac{dv}{dt} = f(v) - v$$

$$\frac{dv}{dt} = \frac{f(v) - v}{t}$$

$$\frac{dv}{f(v) - v} = \frac{dt}{t} \leftarrow v(t) = \frac{y(t)}{t}$$

$$\frac{dy}{dt} = 1 - \frac{y}{t} + \frac{y^2}{t^2}$$

$$t\frac{dv}{dt} + v = 1 - v + v^2$$

$$t\frac{dv}{dt} = 1 - 2v + v^2$$

$$\frac{dv}{1 - 2v + v^2} = \frac{dt}{t}$$

$$\frac{dv}{1+2v^2+v^2} = \frac{dt}{t}$$

$$\frac{dv}{(1-v)^2} = \frac{dt}{t}$$

$$-(1-v)^{-1} = \ln t + c$$

$$-(1-y/t)^{-1} = \ln t + c$$

linear:

$$\frac{dy}{dt} + a(t)y = f(t) \leftarrow$$

$$\frac{dy}{dt} = -a(t)y + f(t)$$

$$e^{A(t)}$$

$$\frac{d}{dt}[y(t)e^{A(t)}] = \frac{dy}{dt}e^{A(t)} + \frac{dA}{dt}e^{A(t)}y(t)$$

$$= e^{A(t)}\left[\frac{dy}{dt} + \frac{dA}{dt}y(t)\right]$$

let $\frac{dA}{dt} = a(t)$ $= e^{A(t)}\left[\frac{dy}{dt} + a(t)y(t)\right]$

$$e^{A(t)}\left[\frac{dy}{dt} + a(t)y\right] = e^{A(t)}f(t)$$

$$\frac{d}{dt}[e^{A(t)}y(t)] = e^{A(t)}f(t)$$

$$\frac{d}{dt}\left[e^{A(t)}y(t)\right] = e^{A(t)}f(t)$$

$$e^{A(t)}y(t) = \int_{t_0}^{t} dt' \, e^{A(t')}f(t') + C$$

$$y(t) = e^{-A(t)}\int_{t_0}^{t} dt' \, e^{A(t')}f(t') + Ce^{-A(t)}$$

$$e^{A(t)}y(t) - e^{A(t_0)}y(t_0) = \int_{t_0}^{t} dt' \, e^{+A(t')}f(t')$$

$$y(t) = e^{A(t_0)-A(t)}y(t_0) + e^{-A(t)}\int_{t_0}^{t} dt' \, e^{+A(t')}f(t')$$

$$A(t) = \int dt \, a(t)$$

$$\frac{dA}{dt} = a(t)$$

$$\frac{dy}{dt} + \frac{2}{t}y = \frac{3}{t^2}$$

($a(t)$ ← $\frac{2}{t}$; $f(t)$ ← $\frac{3}{t^2}$)

$$\frac{d}{dt}\left[e^{A(t)}y\right]$$

$$A(t) = \int dt \, a(t) = \int dt \, \frac{2}{t} = 2\ln t$$

$$e^{A(t)} = e^{2\ln t} = t^2$$

$$\frac{dy}{dt} + \frac{2}{t}y = \frac{3}{t^2} \leftarrow f(t)$$

$$\frac{d}{dt}\left[e^{A(t)}\, y\right] = \frac{d}{dt}\left[t^2 y\right]$$

$$= t^2 \frac{dy}{dt} + 2t\, y(t)$$

$$= t^2\left[\frac{dy}{dt} + \frac{2}{t} y(t)\right]$$

$$e^{A(t)} = t^2$$

$$t^2\left[\frac{dy}{dt} + \frac{2}{t} y(t)\right] = t^2 \frac{3}{t^2}$$

$$\frac{d}{dt}\left[t^2 y\right] = 3$$